



National Load Despatch Centre
राष्ट्रीय भार प्रेषण केंद्र
GRID CONTROLLER OF INDIA LIMITED
ग्रिड कंट्रोलर ऑफ इंडिया लिमिटेड

(Government of India Enterprise/ भारत सरकार का उद्यम)
B-9, QUTUB INSTITUTIONAL AREA, KATWARIA SARAI, NEW DELHI -110016
बी-9, कुतुब इन्स्टीट्यूशनल एरिया, कटवारिया सराये, न्यू दिल्ली-110016

Ref: GRID-INDIA/NLDC/SO/Daily PSP Report

दिनांक: 22.12.2024

To,

- कार्यकारी निदेशक, पू.क्षे.भा.प्रे.के., 14, गोल्फ क्लब रोड, कोलकाता - 700033
Executive Director, ERLDC, 14 Golf Club Road, Tollygunge, Kolkata, 700033
- कार्यकारी निदेशक, ऊ.क्षे.भा.प्रे.के., 18/ए, शाहीद जीत सिंह सनसनवाल मार्ग, नई दिल्ली - 110016
Executive Director, NRLDC, 18-A, Shaheed Jeet Singh Marg, Katwaria Sarai, New Delhi - 110016
- कार्यकारी निदेशक, प.क्षे.भा.प्रे.के., एफ3, एम आई डी सी क्षेत्र, अंधेरी, मुंबई - 400093
Executive Director, WRLDC, F-3, M.I.D.C. Area, Marol, Andheri (East), Mumbai-400093
- कार्यकारी निदेशक, ऊ.पू.क्षे.भा.प्रे.के., डोंगतेह, लोअर नोंग्रह, लापालंग, शिलोंग - 793006
Executive Director, NERLDC, Dongteih, Lower Nongrah, Lapalang, Shillong - 793006, Meghalaya
- कार्यकारी निदेशक, द.क्षे.भा.प्रे.के., 29, रेस कोर्स क्रॉस रोड, बंगलुरु - 560009
Executive Director, SRLDC, 29, Race Course Cross Road, Bangalore-560009

Sub: Daily PSP Report for the date 21.12.2024.

महोदय/Dear Sir,

आईईजीसी-2023 की धारा स.-38(1) के प्रावधान के अनुसार, दिनांक 21-दिसंबर-2024 की अखिल भारतीय प्रणाली की दैनिक ग्रिड निष्पादन रिपोर्ट रांभांप्रेके की वेबसाइट पर उपलब्ध है।

As per article 38(1) of the Indian Electricity Grid Code, the daily report pertaining power supply position of All India Power System for the date 21.12.2024, is available at the NLDC website.

धन्यवाद,
Thanks

Report for previous day

Date of Reporting: 22-Dec-2024

A. Power Supply Position at All India and Regional level

	NR	WR	SR	ER	NER	TOTAL
Demand Met during Evening Peak hrs(MW) (at 19:00 hrs; from RLDCs)	58325	63166	44732	21241	2747	190211
Peak Shortage (MW)	0	483	0	0	0	483
Energy Met (MU)	1249	1521	1127	428	50	4374
Hydro Gen (MU)	113	58	93	17	13	295
Wind Gen (MU)	26	98	17	-	-	141
Solar Gen (MU)*	113.07	78.44	104.87	2.41	0.68	299
Energy Shortage (MU)	2.25	1.58	0.00	0.04	0.00	3.87
Maximum Demand Met During the Day (MW) (From NLDC SCADA)	66927	76378	54267	21888	2915	218790
Time Of Maximum Demand Met	10:15	10:28	08:07	18:00	17:20	09:58

B. Frequency Profile (%)

Region	FVI	< 49.7	49.7 - 49.8	49.8 - 49.9	< 49.9	49.9 - 50.05	> 50.05
All India	0.042	0.15	1.10	7.64	8.89	78.85	12.27

C. Power Supply Position in States

Region	States	Max.Demand Met during the day (MW)	Shortage during maximum Demand (MW)	Energy Met (MU)	Drawal Schedule (MU)	OD(+)/UD(-) (MU)	Max OD (MW)	Energy Shortage (MU)
NR	Punjab	9199	0	162.1	68.9	-2.8	322	0.00
	Haryana	9049	0	166.1	110.7	-1.5	605	0.00
	Rajasthan	17727	0	330.6	108.2	-1.1	861	0.00
	Delhi	4444	0	77.0	68.0	0.0	418	0.00
	UP	19897	0	355.7	99.1	-2.0	680	0.00
	Uttarakhand	2483	0	44.9	34.6	0.5	124	0.00
	HP	2194	0	40.9	34.5	-0.9	142	0.00
	J&K(UT) & Ladakh(UT)	3090	0	63.2	59.1	-0.8	739	2.25
	Chandigarh	249	0	4.2	4.4	-0.3	38	0.00
WR	Railways_NR ISTS	198	0	4.1	3.7	0.4	34	0.00
	Chhattisgarh	4991	0	103.9	44.4	0.6	322	0.00
	Gujarat	22360	0	420.0	185.3	-1.3	913	0.00
	MP	18451	171	345.4	220.8	-1.6	917	1.58
	Maharashtra	28234	0	575.7	209.0	0.6	520	0.00
	Goa	683	0	13.9	13.3	0.0	58	0.00
	DNHDDPDCL	1322	0	30.2	30.1	0.1	193	0.00
	AMNSIL	738	0	16.7	9.0	0.0	253	0.00
	BALCO	529	0	12.6	12.6	0.0	48	0.00
SR	RIL JAMNAGAR	101	0	2.4	2.3	0.1	0	0.00
	Andhra Pradesh	9586	0	194.3	57.2	-0.8	908	0.00
	Telangana	13839	0	266.5	164.3	0.7	1221	0.00
	Karnataka	14901	0	263.7	80.8	0.2	802	0.00
	Kerala	4333	0	85.6	65.8	1.5	328	0.00
	Tamil Nadu	15547	0	307.6	195.7	1.1	957	0.00
	Puducherry	420	0	9.3	8.9	-0.1	40	0.00
ER	Bihar	5029	0	92.9	83.1	-2.3	235	0.04
	DVC	3184	0	68.9	-44.7	0.8	249	0.00
	Jharkhand	1983	0	38.5	26.9	0.0	346	0.00
	Odisha	4559	0	85.0	16.1	-1.7	332	0.00
	West Bengal	6974	0	140.5	1.3	-1.6	204	0.00
	Sikkim	116	0	1.8	1.9	-0.1	26	0.00
	Railways_ER ISTS	21	0	0.2	0.2	0.0	0	0.00
NER	Arunachal Pradesh	173	0	3.1	2.9	0.0	17	0.00
	Assam	1545	0	27.1	20.6	0.3	136	0.00
	Manipur	252	0	3.6	3.8	-0.2	46	0.00
	Meghalaya	400	0	6.5	5.7	-0.2	70	0.00
	Mizoram	162	0	2.5	1.9	0.1	28	0.00
	Nagaland	168	0	2.6	2.5	0.0	23	0.00
	Tripura	228	0	4.1	3.2	-0.1	37	0.00

D. Transnational Exchanges (MU) - Import(+ve)/Export(-ve)

	Bhutan	Nepal	Bangladesh	Godda -> Bangladesh
Actual (MU)	-7.2	-5.5	-20.9	-12.4
Day Peak (MW)	-507.3	-357.9	-974.0	-748.3

E. Import/Export by Regions (in MU) - Import(+ve)/Export(-ve); OD(+)/UD(-)

	NR	WR	SR	ER	NER	TOTAL
Schedule(MU)	231.7	-251.5	219.6	-202.2	2.3	0.0
Actual(MU)	227.1	-269.6	240.2	-206.9	-0.1	-9.3
O/D/U/D(MU)	-4.6	-18.2	20.6	-4.7	-2.4	-9.3

F. Generation Outage(MW)

	NR	WR	SR	ER	NER	TOTAL	% Share
Central Sector	4910	8921	6528	4538	368	25264	47
State Sector	5183	13227	8274	1869	171	28723	53
Total	10093	22147	14802	6407	538	53987	100

G. Sourcewise generation (Gross) (MU)

	NR	WR	SR	ER	NER	All India	% Share
Coal	764	1563	604	699	15	3645	76
Lignite	29	10	40	0	0	80	2
Hydro	113	58	93	17	13	295	6
Nuclear	31	66	70	0	0	167	4
Gas, Naptha & Diesel	13	39	5	0	25	82	2
RES (Wind, Solar, Biomass & Others)	165	179	150	6	1	500	10
Total	1115	1915	963	722	54	4768	100

Share of RES in total generation (%)	14.81	9.34	15.54	0.79	1.26	10.49
Share of Non-fossil fuel (Hydro,Nuclear and RES) in total generation(%)	27.76	15.82	32.52	3.18	25.33	20.18

H. All India Demand Diversity Factor

Based on Regional Max Demands	1.016
Based on State Max Demands	1.048

I. All India Peak Demand and shortage at Solar and Non-Solar Hour

	Max Demand Met(MW)	Time	Shortage(MW)
Solar hr	218790	9:58	621
Non-Solar hr	201201	18:29	260

Diversity factor = Sum of regional or state maximum demands / All India maximum demand

**Note: All generation MU figures are gross

***Godda (Jharkhand) -> Bangladesh power exchange is through the radial connection (isolated from Indian Grid)

Solar Hours -> 06:00 to 18:00hrs and rest are Non-Solar Hours

*Source: RLDCs for solar connected to ISTS; SLDCs for embedded solar. Limited visibility of embedded solar data.

Executive Director-NLDC

INTER-REGIONAL EXCHANGES								
Import=(+ve) /Export =(-ve) for NET (MU)								
Date of Reporting: 22-Dec-2024								
Sl No	Voltage Level	Line Details	No. of Circuit	Max Import (MW)	Max Export (MW)	Import (MU)	Export (MU)	NET (MU)
Import/Export of ER (With NR)								
1	HVDC	ALIPURDUAR-AGRA	2	0	0	0.0	0.0	0.0
2	HVDC	PUSAULI B/B	-	0	251	0.0	6.1	-6.1
3	765 kV	GAYA-VARANASI	2	0	980	0.0	15.5	-15.5
4	765 kV	SASARAM-FATEHPUR	1	0	556	0.0	9.3	-9.3
5	765 kV	GAYA-BALIA	1	0	688	0.0	11.5	-11.5
6	400 kV	PUSAULI-VARANASI	1	0	180	0.0	3.6	-3.6
7	400 kV	PUSAULI -ALLAHABAD	1	0	124	0.0	2.5	-2.5
8	400 kV	MUZAFFARPUR-GORAKHPUR	2	0	614	0.0	9.5	-9.5
9	400 kV	PATNA-BALIA	2	0	853	0.0	16.8	-16.8
10	400 kV	NAUBATPUR-BALIA	2	0	533	0.0	5.3	-5.3
11	400 kV	BIHARSHARIFF-BALIA	2	0	199	0.0	4.2	-4.2
12	400 kV	MOTIHARI-GORAKHPUR	2	0	554	0.0	10.2	-10.2
13	400 kV	BIHARSHARIFF-VARANASI	2	0	360	0.0	0.0	0.0
14	220 kV	SAHUPURI-KARAMNASA	1	0	64	0.0	1.1	-1.1
15	132 kV	NAGAR UNTARI-RIHAND	1	0	0	0.1	0.0	0.1
16	132 kV	GARWAH-RIHAND	1	30	0	0.6	0.0	0.6
17	132 kV	KARMANASA-SAHUPURI	1	0	0	0.0	0.0	0.0
18	132 kV	KARMANASA-CHANDAULI	1	0	0	0.0	0.0	0.0
ER-NR						0.7	95.4	-94.7
Import/Export of ER (With WR)								
1	765 kV	JHARSUGUDA-DHARAMJAIGARH	4	109	644	0.0	7.2	-7.2
2	765 kV	NEW RANCHI-DHARAMJAIGARH	2	429	982	0.0	4.7	-4.7
3	765 kV	JHARSUGUDA-DURG	2	0	673	0.0	12.9	-12.9
4	400 kV	JHARSUGUDA-RAIGARH	4	0	696	0.0	10.9	-10.9
5	400 kV	RANCHI-SIPAT	2	0	2	0.0	0.0	0.0
6	220 kV	BUDHIPADAR-RAIGARH	1	9	98	0.0	1.4	-1.4
7	220 kV	BUDHIPADAR-KORBA	2	70	86	0.0	0.0	0.0
ER-WR						0.0	37.1	-37.1
Import/Export of ER (With SR)								
1	HVDC	JEYPORE-GAZUWAKA B/B	2	0	635	0.0	11.7	-11.7
2	HVDC	TALCHER-KOLAR BIPOLE	2	0	1999	0.0	47.4	-47.4
3	765 kV	ANGUL-SRIKAKULAM	2	0	2977	0.0	56.3	-56.3
4	400 kV	TALCHER-I/C	2	0	745	0.0	14.1	-14.1
5	220 kV	BALIMELA-UPPER-SILERRU	1	0	0	0.0	0.0	0.0
ER-SR						0.0	115.3	-115.3
Import/Export of ER (With NER)								
1	400 kV	BINAGURI-BONGAIGAON	2	233	130	2.1	0.5	1.5
2	400 kV	ALIPURDUAR-BONGAIGAON	2	445	129	4.4	0.0	4.4
3	220 kV	ALIPURDUAR-SALAKATI	2	95	26	1.0	0.0	1.0
ER-NER						7.4	0.5	6.9
Import/Export of NER (With NR)								
1	HVDC	BISWANATH CHARIALI-AGRA	2	654	0	7.4	0.0	7.4
NER-NR						7.4	0.0	7.4
Import/Export of WR (With NR)								
1	HVDC	CHAMPA-KURUKSHETRA	2	0	1515	0.0	29.3	-29.3
2	HVDC	VINDHYACHAL B/B	-	147	55	1.9	0.6	1.3
3	HVDC	MUNDRA-MOHINDERGARH	2	0	983	0.0	24.2	-24.2
4	765 kV	GWALIOR-AGRA	2	0	1750	0.0	25.6	-25.6
5	765 kV	GWALIOR-PHAGI	2	0	1695	0.0	25.1	-25.1
6	765 kV	JABALPUR-ORAI	2	0	915	0.0	26.4	-26.4
7	765 kV	GWALIOR-ORAI	1	830	0	14.9	0.0	14.9
8	765 kV	SATNA-ORAI	1	0	1025	0.0	20.6	-20.6
9	765 kV	BANASKANTHA-CHITORGARH	2	955	825	6.0	5.6	0.4
10	765 kV	VINDHYACHAL-VARANASI	2	0	2245	0.0	35.6	-35.6
11	400 kV	ZERDA-KANKROLI	1	192	100	1.3	0.4	0.9
12	400 kV	ZERDA -BHINMAL	1	132	70	0.9	0.3	0.6
13	400 kV	VINDHYACHAL -RIHAND	1	964	0	21.9	0.0	21.9
14	400 kV	RAPP-SHUJALPUR	2	385	296	2.6	0.9	1.7
15	400 kV	NEEMUCH-Chittorgarh	2	431	279	3.5	0.6	2.9
16	220 kV	BHANPURA-RANPUR	1	0	203	0.0	3.4	-3.4
17	220 kV	BHANPURA-MORAK	1	0	30	0.0	2.1	-2.1
18	220 kV	MEHGAON-AURAIYA	1	137	0	1.5	0.0	1.5
19	220 kV	MALANPUR-AURAIYA	1	105	0	0.9	0.0	0.9
20	132 kV	GWALIOR-SAWAI MADHOPUR	1	0	0	0.0	0.0	0.0
21	132 kV	RAJGHAT-LALITPUR	2	0	0	0.0	0.0	0.0
WR-NR						55.3	200.7	-145.3
Import/Export of WR (With SR)								
1	HVDC	BHADRAWATI B/B	-	0	1008	0.0	24.0	-24.0
2	HVDC	RAIGARH-PUGALUR	2	0	5009	0.0	66.1	-66.1
3	765 kV	SOLAPUR-RAICHUR	2	1186	1476	3.9	12.8	-8.8
4	765 kV	WARDHA-NIZAMABAD	2	0	2652	0.0	38.9	-38.9
5	765 kV	WARORA-WARANGAL(NEW)	2	0	2686	0.0	41.8	-41.8
6	400 kV	KOLHAPUR-KUDGI	2	1375	0	16.1	0.0	16.1
7	220 kV	KOLHAPUR-CHIKODI	2	0	0	0.0	0.0	0.0
8	220 kV	PONDA-AMBEWADI	1	0	0	0.0	0.0	0.0
9	220 kV	XELDEM-AMBEWADI	1	0	122	2.3	0.0	2.3
WR-SR						22.3	183.6	-161.3
INTERNATIONAL EXCHANGES								
State	Region	Line Name	Max (MW)	Min (MW)	Avg (MW)	Import(+ve)/Export(-ve) Energy Exchange (MU)		
BHUTAN	ER	400kV MANGDECHHU-ALIPURDUAR 1,2&3 i.e. ALIPURDUAR RECEIPT (from MANGDECHU HEP 4*180MW)	-118	110	-30	-0.71		
	ER	400kV TALA-BINAGURI 1,2,4 (& 400kV MALBASE - BINAGURI) i.e. BINAGURI RECEIPT (from TALA HEP 6*170MW)	-213	180	-55	-1.33		
	ER	220kV CHUKHA-BIRPARA 1&2 (& 220kV MALBASE - BIRPARA) i.e. BIRPARA RECEIPT (from CHUKHA HEP 4*84MW)	-256	-132	-201	-4.82		
	NER	132kV GELEPHU-SALAKATI	-26	-9	-18	-0.43		
	NER	132kV MOTANGA-RANGIA	23	0	4	0.10		
NEPAL	NR	NEPAL IMPORT (FROM UP)	-60	0	0	0.00		
	NR	132kV MAHENDRANAGAR-TANAKPUR(NHPC)	-60	0	-38	-0.91		
	ER	NEPAL IMPORT (FROM BIHAR)	-68	-41	-56	-1.34		
	ER	400kV DHALKEBAR-MUZAFFARPUR 1&2	-290	-4	-135	-3.24		
BANGLADESH	ER	BHERAMARA B/B HVDC (B'DESH)	-926	-542	-838	-20.11		
	ER (Isolated from Indian Grid)	400kV GODDA_TPS-RAHANPUR (B'DESH) D/C	-748	-409	-516	-12.40		
	NER	132kV COMILLA-SURAJMANI NAGAR 1&2	-48	0	-34	-0.82		

CROSS BORDER EXCHANGE SCHEDULE

Date of Reporting: 22-Dec-2024

Export From India (in MU)

Country	GNA (ISGS/PPA)	T-GNA							TOTAL	
		BILATERAL TOTAL	COLLECTIVE							
			IDAM			RTM				
			IEX	PXIL	HPX	IEX	PXIL	HPX		
Bhutan	0.00	0.00	8.44	0.00	0.00	0.00	0.00	0.00	8.44	
Nepal	1.33	0.00	2.34	0.00	0.00	0.00	1.50	0.00	0.00	5.17
Bangladesh	21.04	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.04
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Export	22.37	0.00	10.78	0.00	0.00	0.00	1.50	0.00	0.00	34.65

Import by India(in MU)

Country	GNA (ISGA/PPA)	T-GNA							TOTAL
		BILATERAL TOTAL	COLLECTIVE						
			IDAM			RTM			
			IEX	PXIL	HPX	IEX	PXIL	HPX	
Bhutan	0.60	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.60
Nepal	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Bangladesh	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Import	0.60	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.60

Net from India(in MU)

-ve : Export / +ve : Import

Country	GNA (ISGS/PPA)	T-GNA							TOTAL	
		BILATERAL TOTAL	COLLECTIVE							
			IDAM			RTM				
			IEX	PXIL	HPX	IEX	PXIL	HPX		
Bhutan	0.60	0.00	-8.44	0.00	0.00	0.00	0.00	0.00	-7.84	
Nepal	-1.33	0.00	-2.34	0.00	0.00	0.00	-1.50	0.00	0.00	-5.17
Bangladesh	-21.04	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-21.04
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Net	-21.77	0.00	-10.78	0.00	0.00	0.00	-1.50	0.00	0.00	-34.05

